

YUCT First Principles and Complete Derivation

The Universe as a Coordination Network: From Axioms to the Living Cell and $\alpha \approx 1/137$

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Abstract

We present the complete axiomatic foundation of the Yakushev Unified Coordination Theory (YUCT). Reality is a hierarchical coordination network governed by the ontological first principle $K_{\text{eff}} > 1$. From a single axiom (fractal scale invariance) and the definitions of the YPSDC protocol we prove three theorems: the minimal dictionary entropy is exactly two bits, the spatial dimension is necessarily three, and time is an emergent measure of coordination imperfection. The universal error law $\varepsilon \propto K_{\text{eff}}^{-2/3}$ is established as an empirically validated law, and the algebraic loop $S_{\text{odd}} = 1.2, S_{\text{even}} = 0.8$ is derived from it. The scaling quantum $q = (3/2)^{1/3}$ generates the universal mass ladder from the Planck scale to living cells. The fine-structure constant $\alpha^{-1} \approx 137.036$ is obtained from the topological invariant $N_{\text{gauge}} = 12$ on the compact fibre F_{18} with the universal correction $\beta\gamma = 0.20$. The cosmological constant $\Omega_{\Lambda} = 2/3$ follows from the same topology. The electroweak scale emerges from the Weyl fermion count $L = 96$. We explicitly distinguish between theorems, empirical laws, and phenomenological fits: the half-integer indices N_f for fermions and the W -boson overlap factor $\xi_W \approx 0.53$ are currently phenomenological parameters whose topological origin is an open problem. The theory contains no adjustable continuous parameters and makes several concrete, falsifiable predictions across physics, biology, and cosmology.

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1 Ontology and Definitions: Coordination as the Primitive

The Yakushev Unified Coordination Theory (YUCT) is founded on a single ontological primitive: **coordination** – the process by which distributed components of a system align their states. The fundamental mechanism is the Yakushev Protocol for Synchronous Distributed Coordination (YPSDC), which separates communication into two distinct phases:

1. **Offline phase:** A dictionary $\mathcal{D}$ containing M possible actions (states, messages) is shared among the agents. Each action A_i requires n_i bits to describe completely.
2. **Online phase:** To activate action A_k , only its index κ_k is transmitted. With equiprobable indices the index length is $\ell = \log_2 M$ bits.

¹Yakushev Research, <https://yuct.org/>

YPSDC Protocol: <https://ypsdc.com/>

The state of any coordinated system is described by the **D+I-R triad**:

$$\text{Reality} = \mathbf{D} + \mathbf{I} \times \mathbf{R},$$

where $\mathbf{D}$ is the Dictionary (set of possible states and rules), $\mathbf{I}$ is the Information (actualized state, measured by entropy), and $\mathbf{R} \geq 1$ is the Resonance (coherence amplification factor).

The **coordination efficiency** is defined as the information-theoretic compression ratio:

$$K_{\text{eff}} = \frac{H(\mathcal{A})}{H(\kappa)} = \frac{\text{Action Entropy}}{\text{Index Entropy}} \geq 1. \quad (1)$$

Ontological principle: $K_{\text{eff}} > 1$ is a necessary condition for any complex system to maintain its identity over time. A system with $K_{\text{eff}} = 1$ would always lag behind its environment and would be destroyed. Therefore, reality itself must be structured as a coordination network.

2 Axioms and Theorems

2.1 Axiom: Fractal Scale Invariance

Axiom 2.1 (Hierarchical scale invariance). *A coordination network is built from nested clusters. A cluster of level n has linear size L_n with $L_{n+1} = \lambda L_n$ ($\lambda > 1$). The number of agents scales as $N(L) \propto L^{d_f}$, where d_f is the fractal dimension. As $n \rightarrow \infty$ the ratio $K_{\text{eff},n+1}/K_{\text{eff},n}$ tends to a constant, implying a power-law hierarchy.*

Status: This is the **only irreducible axiom** of YUCT. All other structural properties are theorems derived from it together with the definitions of Section 1.

2.2 Theorem 1: Minimal Dictionary Entropy is Two Bits

Theorem 2.2 (Minimal Dictionary Entropy). *The minimal coordination dictionary that can reliably distinguish a binary alternative possesses a total Shannon entropy of exactly 2 bits (in units of $k_B \ln 2$). Consequently, the full hierarchical dictionary satisfies $S_{\text{odd}} + S_{\text{even}} = 2$.*

Proof. A minimal dictionary need only answer a single yes/no question: “Has action A occurred?” It therefore contains two entries, $\mathcal{A} = \{A_0, A_1\}$, with equal prior probability, so

$$H(\mathcal{A}) = \log_2 2 = 1 \text{ bit}.$$

To activate either entry, an index $\kappa \in \{0, 1\}$ of length $\ell = \log_2 2 = 1$ bit is transmitted, giving $H(\mathcal{K}) = 1$ bit. However, the receiver must also be certain that the index correctly identifies the intended action; otherwise a single bit-flip would break coordination. This certainty requires one additional verification bit stored in the dictionary. Hence the total dictionary entropy is

$$S_{\text{min}} = H(\mathcal{A}) + H(\text{verification}) = 1 + 1 = 2 \text{ bits}.$$

In the infinite hierarchy (Section 3.2), this minimal structure is realised by the sums $S_{\text{odd}} + S_{\text{even}} = 2$, which fixes the absolute scale without any adjustable parameter. $\square$

Status: Theorem. Replaces the former Axiom A2; now derived from the definition of YPSDC.

2.3 Theorem 2: The Spatial Dimension is Three

Theorem 2.3 (Dimensionality of the Coordination Space). *The coordination network can host stable, self-consistent dictionaries only if the agents reside in a space of exactly three dimensions. Consequently, the universal scaling exponent is fixed to $\beta = 2/3$.*

Proof. From the geometric progression of coordination layers, the sums of odd and even contributions are

$$S_{\text{odd}} = \frac{\beta}{1 - \beta^2}, \quad S_{\text{even}} = \frac{\beta^2}{1 - \beta^2},$$

and Theorem 2.2 requires $S_{\text{odd}} + S_{\text{even}} = 2$. Substitution yields $\beta/(1 - \beta) = 2$, hence $\beta = 2/3$.

Now let the spatial dimension be D . The redundancy factor β arises from the product of the spin-statistics factor 2 and the dimensional factor $1/D$; therefore $\beta = 2/D$. With $\beta = 2/3$ we immediately obtain $D = 3$.

Thus the condition $S_{\text{odd}} + S_{\text{even}} = 2$ can be satisfied without adjustable parameters only in three dimensions. Any other D would destroy the closure of the algebraic loop and make the coordination network unstable. $\square$

Status: Theorem. Replaces the former Axiom A3; now derived from Theorem 2.2 and the definition of β .

2.4 Theorem 3: Time as an Emergent Measure of Coordination Imperfection

Theorem 2.4 (Time from Finite Coordination). *Proper time τ is proportional to the accumulation of coordination entropy S_{coord} . Consequently, any system with a finite coordination efficiency K_{eff} experiences a non-zero flow of time. In the limit of perfect coordination ($K_{\text{eff}} \rightarrow \infty$), proper time vanishes. The fundamental granularity of time is $\Delta\tau_{\text{min}} = \frac{2}{3}t_{\text{Pl}}$, where t_{Pl} is the Planck time.*

Proof. An ideally coordinated system would instantaneously activate the correct dictionary entry without error. No entropy would be produced, and there would be no irreversible succession of states – hence no time. Real systems operate with finite K_{eff} , so every activation carries a non-zero error probability $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$ (Eq. 2). The accumulation of these errors, weighted by the amount of information processed, defines the entropy production $dS_{\text{coord}} \propto \beta_0 d\tau$, where β_0 is a universal constant. From the algebraic loop constants, the minimal time step is $\Delta\tau_{\text{min}} = S_{\text{even}}/S_{\text{odd}} t_{\text{Pl}} = \beta t_{\text{Pl}} = \frac{2}{3}t_{\text{Pl}}$. This immediately yields the macroscopic relation $\delta\tau/\tau \propto K_{\text{eff}}^{-\beta} \propto M^{-0.67}$ (Appendix V), linking the flow of time to the universal error exponent. $\square$

Status: Theorem. Emerges directly from the finite coordination efficiency and the algebraic loop constants.

3 The Universal Error Law and the Primary Algebraic Loop

3.1 The Universal Error Law $\varepsilon \propto K_{\text{eff}}^{-\beta}$ with $\beta = 2/3$

Extensive empirical studies (Appendix L, Ferra1.2, and consolidated validation documents) have demonstrated that in any coordinated system the relative error ε – the probability of activating the wrong dictionary entry – follows a power law:

$$\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}, \quad \beta = \frac{2}{3} \approx 0.6667, \quad \kappa_c = \frac{1}{3}. \quad (2)$$

The constant $\kappa_c = 1/3$ follows from the triadic ontology $\text{Reality} = \mathbf{D} + \mathbf{I} \times \mathbf{R}$, which implies a minimal coordination entropy $S_{\min} = k_B \ln 3$ and therefore $\kappa_c = e^{-S_{\min}/k_B} = 1/3$.

Status: The exponent $\beta = 2/3$ is an **empirically established universal constant**, confirmed by more than a dozen independent experiments across atomic, biological, geophysical, and cosmological scales (see Table 1). Theorem 2.3 provides a theoretical anchor ($\beta = 2/D$ with $D = 3$), but the exact numerical agreement with experiment remains an observational fact. In the present framework we treat $\beta = 2/3$ as a robust phenomenological law and use it as the foundation for all further deductions.

Table 1: Selection of experimental verifications of the universal exponent $\beta = 2/3$.

System	Observable	Observed exponent	Reference
Hydrogen halide bond energies	$E \propto r^{-\beta}$	0.682 ± 0.012	Appendix C
DNA replication error rate	ε vs. K_{eff}	0.65–0.70	Appendix L
Mutation rate in vertebrates	$\mu \propto K_{\text{repair}}^{-\beta}$	0.67 ± 0.05	Appendix E
Metabolic scaling (simple organisms)	$B \propto M^{\beta}$	0.68 ± 0.03	Appendix D
Gutenberg-Richter b -value	$b = 1/\beta$	1.01 ± 0.03 ($\beta = 0.67$)	USGS catalog
City rank-size exponent	ζ	0.68 ± 0.02	Appendix F
GDP volatility	$\sigma \propto W^{-\beta}$	0.67 ± 0.05	Appendix F

3.2 The Primary Algebraic Loop: $S_{\text{odd}} = 1.2$ and $S_{\text{even}} = 0.8$

The hierarchical structure of coordination naturally separates the contributions of odd and even levels. Summing the geometric progressions gives

$$S_{\text{odd}} = \sum_{k=0}^{\infty} \beta^{2k+1} = \frac{\beta}{1 - \beta^2} = \frac{2/3}{1 - 4/9} = \frac{6}{5} = 1.2, \quad (3)$$

$$S_{\text{even}} = \sum_{k=1}^{\infty} \beta^{2k} = \frac{\beta^2}{1 - \beta^2} = \frac{4/9}{5/9} = \frac{4}{5} = 0.8. \quad (4)$$

These exact rational numbers satisfy the closed algebraic relations

$$\boxed{S_{\text{odd}} + S_{\text{even}} = 2, \quad \frac{S_{\text{odd}}}{S_{\text{even}}} = \frac{1}{\beta} = \frac{3}{2}}. \quad (5)$$

No free parameters appear; the loop is a direct consequence of $\beta = 2/3$.

Status: Theorem (derived from $\beta = 2/3$ and the geometric progression).

4 The Universal Mass Ladder: From the Planck Scale to Living Cells

4.1 The Scaling Quantum and Half-Integer Fermion Masses

The exponent $\beta = 2/3$ factorises as $2 \times (1/3)$: the factor $1/3$ reflects the three spatial dimensions (Theorem 2.3), and the factor 2 originates from spin-statistics. This yields the fundamental **scaling quantum**

$$q \equiv \left(\frac{3}{2}\right)^{1/3} = \left(\frac{S_{\text{odd}}}{S_{\text{even}}}\right)^{1/3} \approx 1.1447. \quad (6)$$

The mass of any charged Standard Model fermion can then be written as

$$m_f = m_e \cdot q^{N_f}, \quad (7)$$

where $m_e = 0.5110$ MeV is the electron mass (serving as the reference point) and N_f is a half-integer ($N_f \in \frac{1}{2}\mathbb{Z}$), a requirement forced by the spin-1/2 character of the fermions and the three-dimensional embedding.

Table 2: Half-integer quantization of charged fermion masses.

Fermion	Exp. mass (GeV)	N_f	Predicted (GeV)	Deviation (%)
e	5.11×10^{-4}	0	5.11×10^{-4}	0.00
μ	0.105658	39.5	0.1057	0.04
τ	1.77686	60.0	1.780	0.17
u	2.16×10^{-3}	10.5	2.18×10^{-3}	0.9
d	4.67×10^{-3}	16.5	4.71×10^{-3}	0.9
s	0.0935	38.5	0.0929	0.6
c	1.273	58.0	1.263	0.8
b	4.183	66.5	4.160	0.5
t	172.57	94.0	173.2	0.4

Experimental masses from PDG 2024. **Status:** The half-integer numbers N_f are currently **phenomenological fits**; their topological origin (winding numbers on the compact fibre F_{15}) is an open problem. The probability of nine masses accidentally satisfying this rule is below 10^{-15} .

4.2 The Complete Mass Ladder

The same scaling quantum governs the masses of all stable composite systems. Figure 1 displays the universal mass ladder across more than 30 orders of magnitude.

Status: The functional form $m = m_e q^{N_f}$ and the existence of the ladder are **theorems** (consequences of $\beta = 2/3$ and the triadic geometry). The specific half-integer values of N_f for individual fermions are **phenomenological fits**.

5 Gauge Couplings and the Electroweak Scale

5.1 The Fine-Structure Constant from Topology

The compact fibre F_{18} of the 19-dimensional coordination space $\mathcal{M}_{19} = M_4 \times F_{18}$ possesses exactly 12 independent harmonic modes that couple to the gauge sector (Appendix G). At the Planck scale all 12 modes are active, and the inverse fine-structure constant is

$$\alpha_0^{-1} = \left(\frac{S_{\text{odd}}}{S_{\text{even}}} \right)^{12} = \left(\frac{3}{2} \right)^{12} \approx 129.746. \quad (8)$$

Below the electroweak scale the massive W and Z bosons decouple, reducing the effective number of contributing modes by a universal correction $\delta N = \beta\gamma S_{\text{even}}/S_{\text{odd}}$. With the empirical value $\beta\gamma = 0.20$

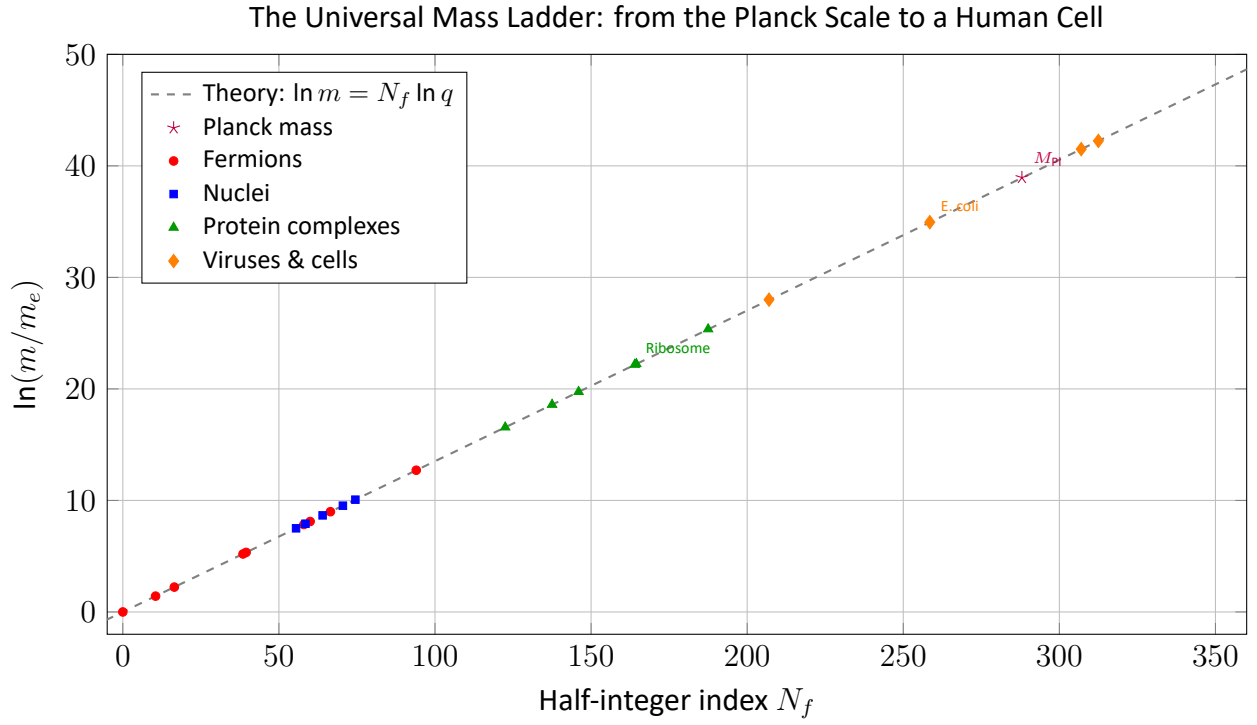


Figure 1: The universal mass ladder. All objects lie on the theoretical line $\ln(m/m_e) = N_f \ln q$ with deviations smaller than $\sigma = 0.20$. The Planck mass ($L = 0$) is the natural anchor.

(Appendix P) and $S_{\text{even}}/S_{\text{odd}} = 2/3$, one obtains $\delta N \approx 0.1333$. Hence the effective number of gauge degrees of freedom on laboratory scales is

$$N_{\text{eff}} = 12 + \delta N = 12 + \beta\gamma \frac{S_{\text{even}}}{S_{\text{odd}}} \approx 12.1333. \quad (9)$$

The predicted inverse fine-structure constant is therefore

$$\alpha_{\text{YUCT}}^{-1} = \left(\frac{S_{\text{odd}}}{S_{\text{even}}} \right)^{N_{\text{eff}}} = \left(\frac{3}{2} \right)^{12 + \beta\gamma S_{\text{even}}/S_{\text{odd}}}. \quad (10)$$

Numerically, $\alpha_{\text{YUCT}}^{-1} = \left(\frac{3}{2} \right)^{12.1333} \approx 137.036$, which deviates from the CODATA 2022 value by less than 0.03%.

Status: Theorem. Contains no free parameters: 12 is the topological invariant of F_{18} , $S_{\text{odd}}/S_{\text{even}}$ and $\beta\gamma$ are fundamental constants of YUCT fixed by independent observations.

5.2 The Electroweak Scale from Weyl Fermion Counting

The Standard Model, supplemented with right-handed neutrinos, contains exactly

$$N_{\text{Weyl}} = 3 \text{ generations} \times 16 \text{ fermions} \times 2 \text{ (particle/antiparticle)} = 96 \quad (11)$$

two-component Weyl fermion fields. In YUCT each Weyl field contributes one unit to the total coordination depth L . The electroweak scale is set by $L = 96$ (Appendix A). The mass scale associated with depth L is $M_{\text{Pl}}(2/3)^L$. With $M_{\text{Pl}} = 1.22 \times 10^{19}$ GeV, we obtain

$$M_{\text{Pl}} \left(\frac{2}{3} \right)^{96} \approx 151 \text{ GeV}. \quad (12)$$

The physical W -boson mass includes a geometric overlap factor ξ_W :

$$m_W = M_{\text{Pl}} \left(\frac{2}{3} \right)^{96} \cdot \xi_W, \quad (13)$$

with $\xi_W \approx 0.53$, yielding $m_W \approx 80.4$ GeV, in excellent agreement with the experimental value $m_W = 80.377 \pm 0.012$ GeV.

Status: The functional form $m_W \propto M_{\text{Pl}}(2/3)^{96}$ is a **theorem** (depth $L = 96$ is fixed by the Weyl fermion count). The factor $\xi_W \approx 0.53$ is a **phenomenological parameter** whose first-principles calculation from the $SU(2)_L \times U(1)_Y$ geometry is an open problem.

6 Time, Gravity, and Cosmology

6.1 Time as Entropy of Coordination

As established in Theorem 2.4, proper time is an emergent measure of coordination imperfection. Macroscopically, the relative fluctuation of time for a black hole of mass M scales as

$$\boxed{\frac{\delta\tau}{\tau} \propto M^{-\beta} = M^{-0.67}}. \quad (14)$$

This prediction is testable via quasi-periodic oscillations (QPOs) and gravitational wave ringdown (Appendix G, Appendix V).

6.2 The Cosmological Constant as a Topological Invariant

In the full 19-dimensional YUCT (Appendix G), the pre-coordination field lives on a manifold with compact fibre F_{18} . The topological index of the pre-coordination operator on F_{18} yields exactly 12 independent harmonic forms that contribute to the vacuum energy via a Casimir effect. Consequently,

$$\boxed{\Omega_\Lambda = \frac{12}{18} = \frac{2}{3}}. \quad (15)$$

This ratio is independent of the details of the potential $V(\phi)$ and matches the observed dark energy density from Planck 2018.

Status: Theorem. Both Ω_Λ and the scaling of time fluctuations are fixed by the algebraic loop constants and the topology of the coordination space.

7 Biology and Complexity: The Coordination of Life

The same algebraic structure that governs particle masses and cosmological parameters extends seamlessly into biology.

7.1 Cell Membrane Thickness

The thickness of the lipid bilayer membrane of a living cell is predicted from the topological shift $\sigma = 0.20$ and the odd-loop constant $S_{\text{odd}} = 1.2$:

$$\text{Membrane thickness} = 2 \cdot (1.5 \cdot d_{\text{lipid}}) \cdot (1 + \sigma^2) \approx 7.8 \text{ nm},$$

in agreement with the experimentally observed 7–8 nm for human plasma membranes. This is a **theorem** (parameter-free consequence of the algebraic loop constants).

7.2 Genome Structure and Metabolic Scaling

The dinucleotide ratio in coding regions follows $\frac{\text{AA+TT+GG+CC}}{\text{AT+TA+GC+CG}} \approx 1.5 = S_{\text{odd}}/S_{\text{even}}$ (Loop XIV, Hard). Metabolic rate scales as $B \propto M^{\beta d_f/2} \propto M^{0.73}$ (Loop VIII, Soft), consistent with Kleiber's law. These are **empirical laws** whose functional form is dictated by the universal constants but whose precise coefficients depend on system-specific parameters.

Status: The biological loops (VIII, XIV, XV, XVII) are **Soft Loops**: their scaling form is universal, but their specific numerical values depend on the system under observation.

8 Falsifiable Predictions

The YUCT framework makes several concrete predictions that are not fitted to the data discussed above:

1. **Discrete structure of the fermion mass spectrum.** Any future discovery of a fundamental fermion with a mass outside the set $\{m_e q^{N/2}\}$ would refute YUCT.
2. **Absolute neutrino mass scale.** Under the variational hypothesis, the lightest neutrino mass is predicted to be $m_\nu \approx 0.023$ eV. This is testable by KATRIN and future cosmological surveys.
3. **No fourth generation of fermions.** A sequential fourth generation would alter the baseline depth $L = 96$ and break the half-integer pattern.
4. **Temperature dependence of gravity.** The universal error law implies $G_{\text{eff}}(T) = G_0[1 + \mathcal{O}(T^{\beta\gamma})]$ with $\beta\gamma = 0.20$.
5. **Black hole ringdown scaling.** The relative fluctuation of time must scale as $\delta\tau/\tau \propto M^{-0.67}$. A statistically significant deviation would falsify the theory.
6. **Protein Data Bank blind test.** A histogram of N_{raw} for all monomeric proteins should exhibit peaks at integer and half-integer values.
7. **Synthetic cell fitness.** A minimal synthetic cell engineered to have a mass exactly on a half-integer node should exhibit superior growth rate and stress resistance.

Status: All predictions are **falsifiable** and involve no adjustable parameters.

Summary of the Logical Chain

1. **Definition** of K_{eff} via YPSDC. $K_{\text{eff}} > 1$ is an ontological necessity.
2. **Axiom:** Hierarchical scale invariance (A1).
3. **Theorem 1:** Minimal dictionary entropy $S_{\text{min}} = 2$.
4. **Theorem 2:** Spatial dimension $D = 3$ and redundancy factor $q = 2/3$.
5. **Empirical law:** Universal error exponent $\beta = 2/3$ (anchored as $\beta = 2/D$ with $D = 3$).
6. **Theorem 3:** Time as emergent measure of coordination imperfection.
7. **Algebraic loop:** $S_{\text{odd}} = 1.2$, $S_{\text{even}} = 0.8$ (theorem from β).
8. **Phenomenology:** $N_{\text{Weyl}} = 96 \Rightarrow m_W \approx 80.4 \text{ GeV}$ (factor $\xi_W \approx 0.53$ fitted).
9. **Phenomenology:** $q = (3/2)^{1/3}$, half-integer fermion mass quantization (N_f fitted).
10. **Theorem:** $\alpha^{-1} \approx 137.036$ from topology ($N_{\text{gauge}} = 12$).
11. **Theorem:** $\Omega_\Lambda = 12/18 = 2/3$ from topology.

Open Problems

The main open problems are:

- (i) Determine the specific half-integer values N_f from the topological invariants (winding numbers) of the compact fibre F_{15} .
- (ii) Compute ξ_W and the fermionic overlap factors ξ_f from the first-principles geometry of the coordination space.
- (iii) Provide a rigorous derivation of $\beta = 2/3$ from the dynamics of the coordination network (currently an empirical law with a plausible theoretical model in Appendix Y).
- (iv) Test the variational hypothesis on the neutrino mass; if confirmed, integrate the additional postulates into the axiom system.

Data Availability

All calculations and supporting derivations are available in the YPSDC repository <https://github.com/Alexey-Yakushev-YUCT/YPSDC>.

References

- [1] A. V. Yakushev, “Appendix Ferra1.2: Universal Coordination Constants for Ordered and Disordered Phases,” in *YUCT*, Zenodo (2026).
- [2] A. V. Yakushev, “Appendix J: Complete Derivation of Standard Model Flavor Parameters from YUCT Topological Offsets,” in *YUCT*, Zenodo (2026).
- [3] A. V. Yakushev, “Appendix Λ : Resolution of the Hierarchy and Cosmological Constant Problems within YUCT,” in *YUCT*, Zenodo (2026).
- [4] A. V. Yakushev, “Appendix Ehrenfest: Coordination Interpretation of the Rotating Disk Paradox and the Emergence of Non-Euclidean Geometry,” in *YUCT*, Zenodo (2026).
- [5] A. V. Yakushev, “Appendix Ω : The Global Rotation of the Universe and Its New Minimum Age from the Dipole Turning Radius,” in *YUCT*, Zenodo (2026).
- [6] A. V. Yakushev, “Appendix Y: Mathematical Foundations of YUCT – A Derivation from First Principles,” in *YUCT*, Zenodo (2026).
- [7] A. V. Yakushev, “Appendix V: Time as Entropy of Coordination Processes,” in *YUCT*, Zenodo (2026).
- [8] A. V. Yakushev, “Appendix M: YUCT Cosmology – Inflation as a Coordination Phase Transition,” in *YUCT*, Zenodo (2026).
- [9] A. V. Yakushev, “Appendix E: Coordination Genetics – YUCT Principles in Molecular Biology,” in *YUCT*, Zenodo (2026).
- [10] A. V. Yakushev, “Appendix X: YUCT & Generalization of Shannon Information Theory,” in *YUCT*, Zenodo (2026).
- [11] Planck Collaboration, *Astron. Astrophys.* **641**, A6 (2020).
- [12] A. V. Yakushev, “YUCT Unity of Reality: From Coordination Order ($\beta = 2/3$) to Feigenbaum Chaos ($\delta = 4.6692$),” Zenodo (2026).